

CASE STUDY • FRAUD DETECTION MODELING

Predicting Retailer Fraud Before It Becomes a Six-Figure Loss

A validated machine learning model built to catch a rare, high-severity fraud pattern before it turns into a catastrophic loss.

THE CHALLENGE

A Loss That Builds Quietly, Then Hits All at Once

A small number of retailers exploited inventory credit terms in order to gamble with their own product, hoping to secure a big prize. The retailer cashes smaller prize tickets in order to pay their weekly invoice. At a certain point, their invoice becomes too large to manage, and they can't pay their weekly bill. This ends up becoming a large, often six-figure loss, and an inventory write-off.

One of the challenges in detecting and mitigating these instances is the activity appears normal. Gambling and fraud instances are hard to detect unless an analyst is trained in reviewing certain metrics, and the importance of the metrics alone are difficult to quantify.



Receive Inventory

Retailer takes on Scratchers inventory, usually on credit



Play Own Product

Validates and cashes tickets instead of selling, hoping to secure a big prize



Debt Builds Quietly

Balance grows while activity looks normal



Settlement Hits

Invoice comes due that can't be paid

THE APPROACH

Machine Learning Model Built on Past Instances



Tight Scope

Predicts one specific, severe outcome within a 120-day window.



Multi-Source Signal

Combines internal sales and inventory activity with third-party credit data and redemption patterns.



Rare-Event Rigor

Upsampled for training, then tested segmented-out live cases.

HOW IT WORKS

Model Workflow



Data Sources

Sales, inventory, redemption
history & business credit signals



XGBoost Model

Gradient-boosted classifier scores
risk of a severe loss in 120 days



Risk Flag

High-risk accounts flagged for analyst
review and analytical deep dive

THE RESULT

What the Model Delivered

100%

Confirmed fraud cases
correctly flagged in testing

~85%

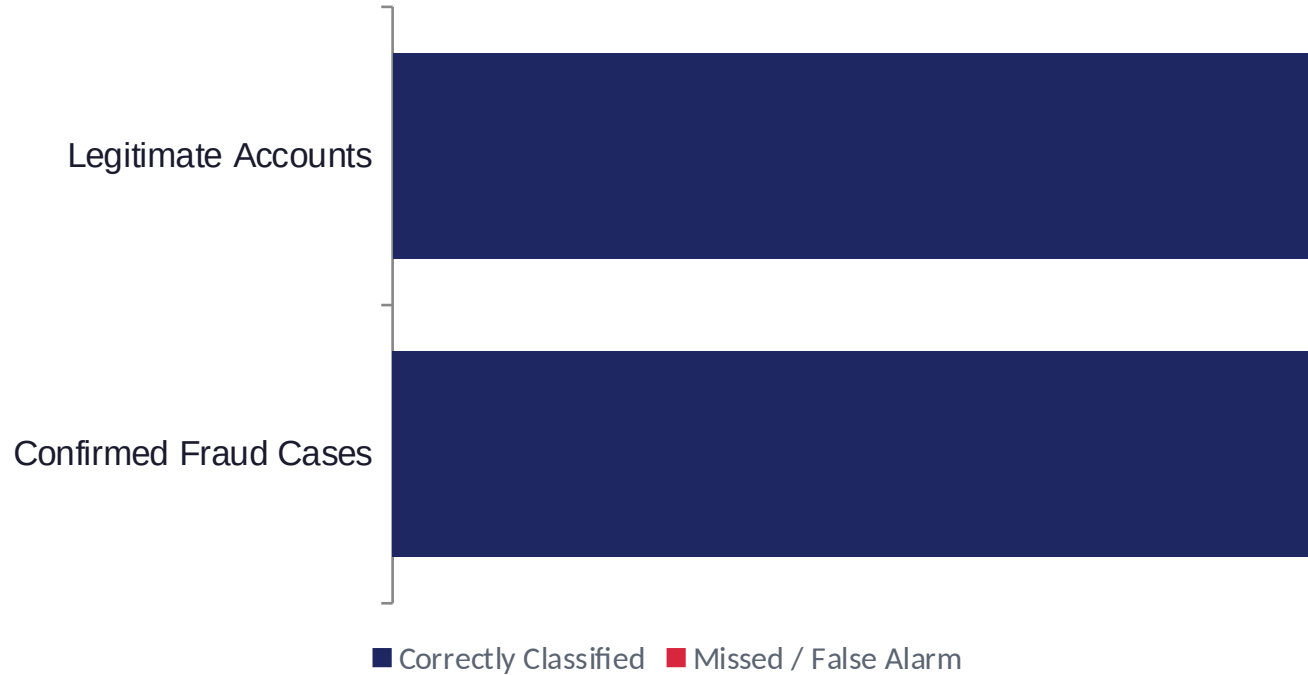
Reduction in this loss
pattern already achieved

\$2.1M

Additional exposure preventable
over the next 5 years

MODEL PRECISION

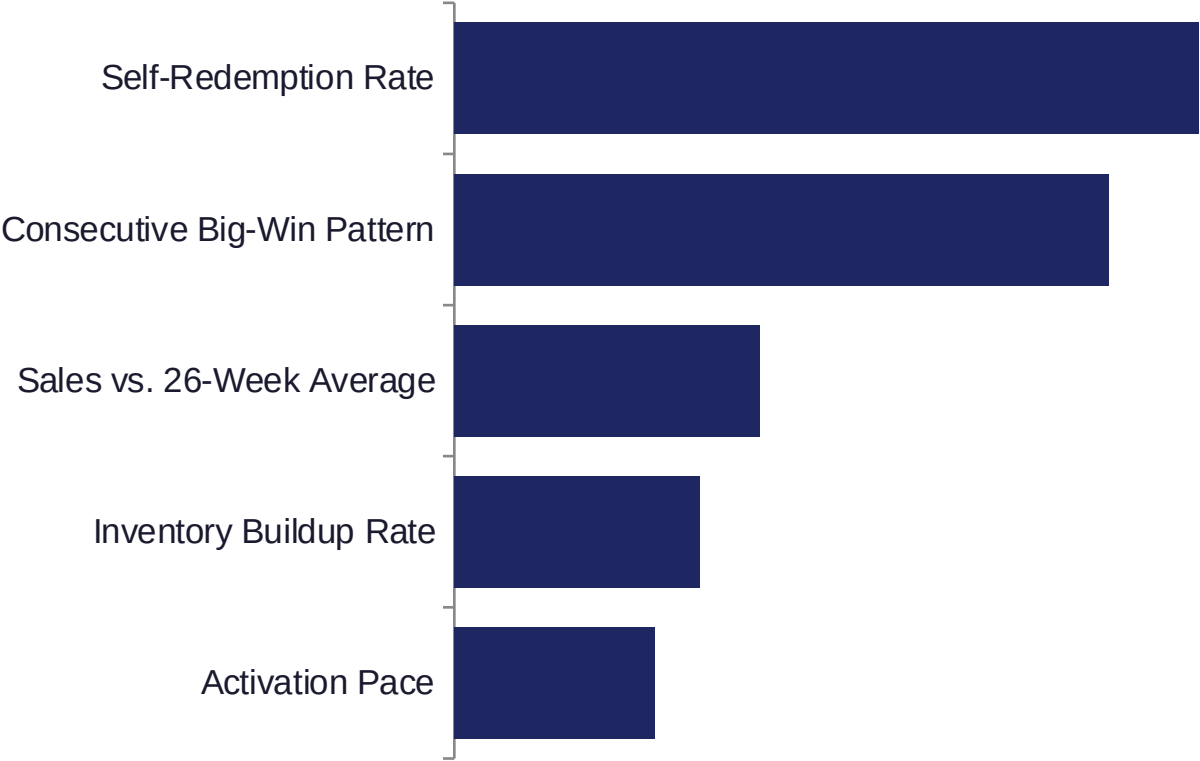
Model Precision – Testing and Training



On a held-out test set the model never trained on, it missed zero confirmed fraud cases and raised only a single false alarm among dozens of legitimate accounts.

WHAT THE MODEL LEARNED

Variable Importance



Retailers potentially committing fraud have several tell-tale signs: cashing an unusually high share of Scratchers tickets they've sold, and validating tickets in large, suspicious blocks. These patterns are almost never seen in ordinary consumer purchasing.

The importance of these data points would not have been uncovered without advanced modeling.

WHY IT MATTERS

Early Detection Leads to Preventing Risk and Saving Retailers



Small Losses Don't Stay Small

Left undetected, fraud and embezzlement instances can develop into large, unmanageable losses in the hundreds of thousands.



Early Detection Can Save the Retailer

With early detection, analysts can apply prevention strategies in order to curb risk and prevent the retailer from incurring a large loss, thus potentially salvaging the lottery/retailer relationship.